Dropout Regularization in Neural Networks Improving Generalisation using MNIST.

Name: Satwik Reddy **Student Id:** 24143945

1. Introduction

In recent years, neural networks have emerged as one of the most popular approaches to machine learning due to their ability to learn complex representations of data. The use of ML models has proven to be highly effective in functions like image classification. Flexible algorithms can also suffer from a problem known as overfitting.

Overfitting refers to a situation wherein a model learns the training pattern of the data including the noise of the input. Consequently, the model does well on training data but poorly on unseen data. It is important regarding real word applications where reliability and generalisation matter a lot.

To solve this issue, many regularisation methods were developed. Dropout is one of the most effective and popular methods. Dropout injections randomness during training by ignoring selected neurons at each training step, allowing the network to learn in a more robust manner.

This paper investigates the operation of dropout using the MNIST dataset, which is a benchmark image dataset. By means of controlled experiments and visual analysis, the study seeks to demonstrate how dropout improves generalisation and reduces overfitting.

2. Background

→Overfitting in Neural Networks:

Overfitting is a frequently occurring issue in machine learning models. A model suffers from overfitting when it overfits the training data such that it takes noise for pattern.
Common indicators of overfitting include.

- The training accuracy is high, but the validation accuracy is low.
- Growing disparity between training and validation results.
- Training loss gets lower while validation loss gets high.

This behaviour shows that the model cannot generalize to new data.

→What is Dropout?

To prevent overfitting of neural networks, dropout is a regularisation technique. Each way, a random subset of neurons are disabled during training. This creates a condition where distinct parts of the network are utilized for training at any point.

This prevents the model from focusing too much on specific neurons and encourages the learning of more general patterns.

→ Intuition Behind Dropout:
The idea of dropout can be understood via a simple analogy. Consider a group of people working on a project. When familiar members always contribute, others will depend on them believing they will always do the needful. When team members are absent at random, everyone is compelled to join in.

Likewise, dropout prevents excessive reliance on any one neuron. Rather, it learns to share the knowledge across several neurons to create a better model.

→ Why Dropout Works
Dropout improves generalisation through several mechanisms:

- **Noise Injection:** Random removal of neurons introduces variability, preventing the model from fitting noise.
- **Model Averaging:** Each training step effectively uses a different network, like ensemble learning.
- **Reduced Co-adaptation:** Neurons cannot depend on specific others, encouraging independent feature learning.

These properties make dropout a powerful tool for improving model performance on unseen data.

3. Methodology

→ Dataset:
The experiments were conducted using the MNIST dataset, which consists of grayscale images of handwritten digits from 0 to 9.

- 60,000 training samples
- 10,000 testing samples
- Image size: 28 × 28 pixels

The dataset is widely used for benchmarking neural network performance and is suitable for analysing overfitting behaviour.

→ Data Preprocessing:
Before training, the following preprocessing steps were applied:

- Pixel values were normalised to the range [0,1]
- Labels were converted into one-hot encoded vectors.
- A validation split of 20% was used.

These steps ensure that the data is properly scaled and suitable for training neural networks.

→ Model Architecture:
We implemented a fully connected neural network with the following architecture.

- Layer that receives input.
- A hidden layer consisting of 128 neurons with ReLU activation function.
- Concealed layer (64 neurons, ReLU activated).
- Output layer: 10 neurons, SoftMax activation.

Two key configurations underwent testing.

- Structure devoid of dropouts.
- A model incorporating dropout at a rate of 0.5.

The impact of varying regularisation strengths was analysed by comparing dropout rates of 0.2 and 0.5.

→ Evaluation Metrics:
To evaluate performance, the following metrics were used:

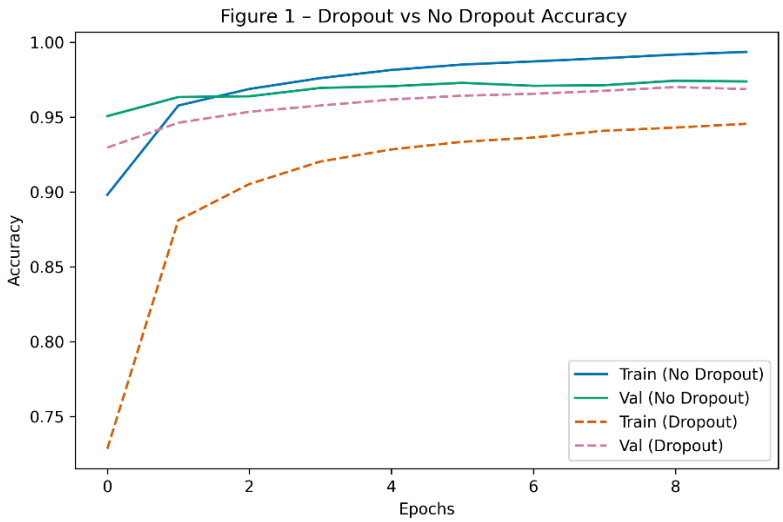
- Accuracy (training and validation)
- Loss (training and validation)
- Overfitting gap (difference between training and validation accuracy)
- Confusion matrix

This combination provides a comprehensive view of both performance and generalisation.

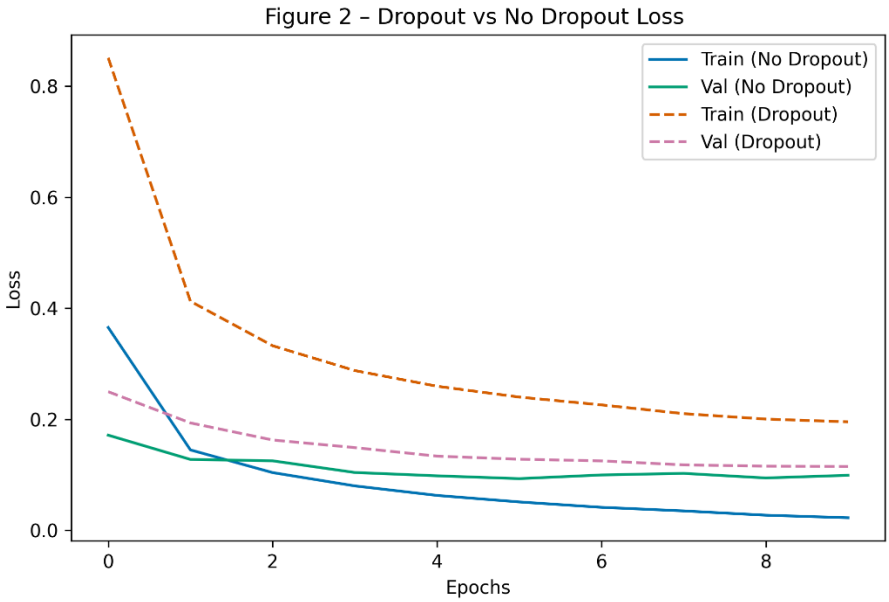
4. Results

→Accuracy Comparison:
The model that does not use dropout exhibits extremely high training accuracy but has a clear gap with validation accuracy. This shows that the model is overfitting.

The dropout model has marginally less training accuracy but slightly improved performance on validation data. This indicates that dropout enables the model to better generalize new data.

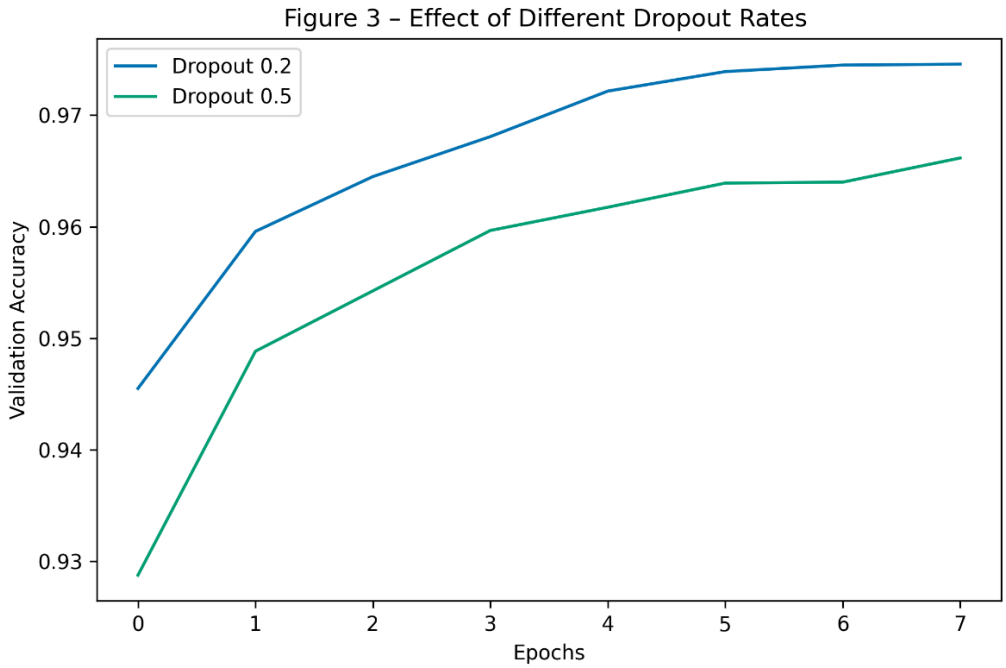


→ Loss Comparison:
These loss curves confirm this. The absence of dropout in the model leads to divergence in training and validation loss. The loss curves of the dropout model are smoother and more stable than the baseline dropout one. That shows it is learning general patterns, rather than noise.

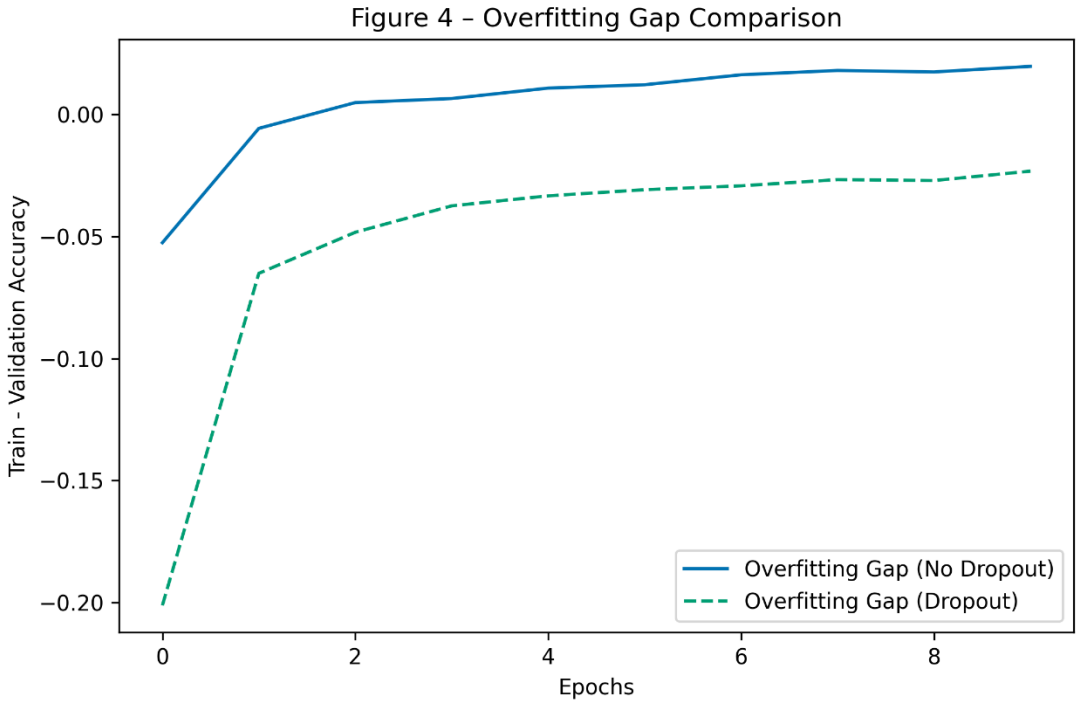


→ Effect of Dropout Rate:
The results show that moderate dropout improves validation accuracy. However, excessive dropout slightly reduces performance, as too many neurons are removed during training.

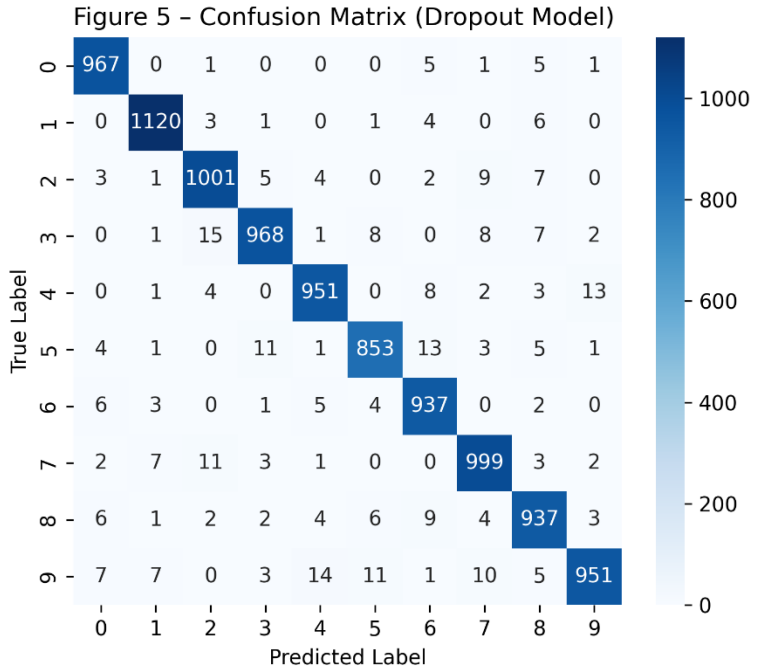
This highlights the importance of selecting an appropriate dropout rate.



→ Overfitting Gap Analysis:
The overfitting gap plot provides a direct measure of generalisation. The model without dropout shows a significantly larger gap, confirming overfitting. The dropout model reduces this gap, demonstrating its effectiveness.



→Confusion Matrix:
The confusion matrix shows that most predictions are correct, with impressive performance across all digit classes. Minor misclassifications occur between similar digits, which is expected.



5. Discussion
The results clearly show that dropout helps in reducing overfitting and increasing generalisation. A model that does not have dropout learns quickly but overfits quickly to the training data. In contrast, the dropout model learns slower and produces more stable predictions.

The model is encouraged to learn more distributed representations through dropout, making it less reliant on individual neurons. Such procedures help foster better robustness and stability.

Also, dropout can be interpreted as an implicit form of ensemble, with many subnetworks trained simultaneously. This is why performance enhancement does not lead to a large cost increase.

6. Critical Analysis
An important finding from this study is that greater accuracy during training may not causally result in better performance. The dropout-free model significantly increases training accuracy while decreasing validation accuracy.

It emphasizes the need for using different metrics for evaluation and focusing on generalisation and not memorisation.

Further, the dropout needs a careful tuning, as the experiment shows. If not, enough dropout occurs, overfitting is not avoided, whereas excessive dropout decreases capacity.

7. Limitations
- Only simple neural network architectures were evaluated.
- Limited range of dropout rates explored.
- MNIST dataset is simple.

- Computational cost was not analysed.

8. Conclusion

- To mitigate the chances of overfitting of a model, dropout layer is commonly used.
- It prevents reliance on specific neurons and improves generalization.
- Models not utilizing dropout usually obtain high training accuracy but show low performance on unseen data.
- When you dropout nodes from a neural network the model takes a more robust decision.
- The most appropriate dropout rate gives regularisation and learning capacity the best trade-off.
- Dropout with a probability close to one would limit the information flowing through the model.
- Dropout enables models to learn features from different distributions which results in the model being robust.
- Visual analysis of accuracy curves, loss curves and overfitting gap indicates its effectiveness.
- Dropout can be seen as an implicit ensemble method that improves model stability.
- In conclusion, dropout improves neural network's reliability and performance in real world situations.

9. Future Work

- Dropout in convolutional neural networks
- Adaptive dropout techniques
- Larger and more complex datasets
- Combining dropout with batch normalisation

10. References

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